Copernicus Global Land Operations
”Vegetation and Energy”
“CGLOPS-1”
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PRODUCT USER MANUAL

DRY MATTER PRODUCTIVITY (DMP)
GROSS DRY MATTER PRODUCTIVITY (GDMP)
NET PRIMARY PRODUCTION (NPP)
GROSS PRIMARY PRODUCTION (GPP)
VERSION 2.0

Issue I1.00

Organization name of lead contractor for this deliverable: VITO

Book Captain:	Dominique De Munck
Contributing Authors:	Carolien Toté
	Sarah Gebruers
	Else Swinnen

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TABLE OF CONTENTS

Executive Summary *11*

1 Background of the document *12*

1.1 Scope and Objectives **12**

1.2 Content of the document **13**

1.3 Related documents **13**

1.3.1 Applicable documents 13

1.3.2 Input 13

1.3.3 External reference 14

2 Algorithm *15*

2.1 Introduction **15**

2.2 Retrieval Methodology **16**

2.2.1 Background 16

2.2.2 Input data 16

2.2.3 Processing steps 17

2.2.4 Near Real Time retrieval of the DMP (NPP) 300m Version 2.0 21

2.3 Differences with the previous version **22**

2.4 Limitations **23**

3 Product Description *25*

3.1 File naming **25**

3.2 File Format **26**

3.3 Product Content **26**

3.3.1 Data Files 26

3.3.2 QFLAG image 27

3.3.3 File and Layer attributes 28

3.4 Product Characteristics **35**

3.4.1 Projection and grid information 35

3.4.2 Spatial information 35

3.4.3 Temporal information 35

3.5 Data Policies **36**

3.6 Access and Contacts **37**

4 Summary of Validation exercise 38

5 Product Usage 39

5.1 Typical use 39

5.2 Advanced usage: how to analyze the data 39

5.2.1 Carbon fluxes 39
5.2.2 Ecosystem and vegetation dynamics 40
5.2.3 Agricultural monitoring 40

6 References 42

7 Annex: User Requirements 45

7.1 Specific technical details and requirements 45

7.2 Further requirements 46

7.2.1 Output product composition 46
7.2.2 Data Structure 46
7.2.3 Data format 46
7.2.4 Uncertainties and Validation 46
7.2.5 Input data 47
7.2.6 Product delivery 47

List of Figures

Figure 1: Global land cover map (ESA CCI epoch 2010) (top) and legend (bottom) derived from ENVISAT MERIS and SPOT-VGT data. 17

Figure 2: fAPAR clamping method. 18

Figure 3: Process flow of GDMP/DMP. Based on meteorological data, a daily GDMPmax is estimated. At the end of each dekad, a Mean Value Composite of these GDMPmax images is calculated. At the same time, a 10-daily fAPAR product is generated. The final $GDMP_{10}$ product is retrieved by the simple multiplication of the latter two images and, then, converted to DMP_{10}. 20

Figure 4: Scheme showing the compositing used for real time estimates 22

Figure 5: Consolidation period example 36

Figure 6 : Global GPP map of the last dekad of June 2024. GPP is expressed in $gC/m^2/day$. 39

Figure 7: Normalized biomass production anomaly (left) and biomass vulnerability index (right) for the year 2025 (ACF, 2025) 41

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List of Tables

Table 1: Summary of key information 12

Table 2: Individual terms in the Monteith variant used for CLMS GDMP/DMP. All terms are expressed on a daily basis. 19

Table 3: Explanation in version numbering and recommendations for using efficiently the products 25

Table 4: Range of values and scaling factors of GDMP/DMP/GPP/NPP. 27

Table 5: Range of values of QFLAG. 28

Table 6: Description of NetCDF file attributes 29

Table 7: Description of NetCDF layers attributes for DMP/GDMP 30

Table 8: Description of netCDF layer attributes for NPP/GPP 32

Table 9: Description of netCDF attributes for coordinate dimensions (latitudes and longitudes) 33

Table 10: Description of netCDF attributes for the grid mapping variable 34

List of Acronyms

ACF	Action Contre la Faim (Action Against Hunger)
AD	Applicable Document
ATBD	Algorithm Theoretical Basis Document
CCI	Climate Change Initiative, an ESA program
CEOS	Committee on Earth Observation System
CF	Climate Forecast
CNDVI	Cumulative NDVI
DM	Dry Matter
DMP	Dry Matter Productivity
DN	Digital number
DTN	Data Transmission Network, United States private company
ε	Efficiency
EBF	Evergreen Broadleaf Forest
EC	European Commission
ECMWF	European Centre for Medium-Range Weather Forecasts
ENVISAT	Environment satellite from ESA, contains MERIS instrument
ESA	European Space Agency
fAPAR	Fraction of absorbed photosynthetically active radiation
GAUL	Global Administrative Unit Layers
GBOV	Ground-Based Observations for Validation
GCOS	The Global Climate Observing System
GDMP	Gross Dry Matter Productivity
GLC	Global Land Cover
GPP	Gross Primary Production
HRES	High Resolution atmospheric model of ECMWF
IGBP	International Geosphere Biosphere Program
J_{AP}	Joule APAR
J_P	Joule PAR
JRC	Joint Research Centre
J_T	Joule total radiation
LAI	Leaf Area Index
LPV	Land Product Validation group of CEOS
LUE	Light use efficiency
MARS	Monitoring Agricultural ResourceS
MARSOP	MARS-operations
MERIS	Medium Resolution Imaging Spectrometer
MODIS	Moderate Resolution Imaging Spectroradiometer
NDVI	Normalized Difference Vegetation Index

NetCDF	Network Common Data Format
NEP	Net Ecosystem Productivity
NGO	Non-Governmental Organization
NPP	Net Primary Production
NRT	Near Real Time
OLCI	Ocean and Land Colour Instrument
PAR	Photosynthetically active radiation
PROBA-V	Vegetation instrument on PROBA platform, Belgian mission
PUM	Product User Manual
QAR	Quality Assessment Report
QFLAG	Quality Flag
R	Radiation
S3	Sentinel-3
SIF	Solar-Induced Fluorescence
SPOT	Système Pour l'Observation de la Terre
T	Temperature
VGT	VEGETATION sensor onboard SPOT4/5
VITO	Flemish Institute for Technological Research
WGS	World Geodetic System

EXECUTIVE SUMMARY

The Copernicus Land Monitoring Service (CLMS) produces a series of qualified bio-geophysical products on the status and evolution of the land surface. The products are used to monitor vegetation, crops, water cycle, energy budget and terrestrial cryosphere. Production and delivery of the parameters take place in a timely manner and are complemented by the constitution of long-term time series.

This document describes the global Dry Matter Productivity (DMP), Gross Dry Matter Productivity (GDMP), Dry Matter Productivity (DMP), Net Primary Production (NPP) and Gross Primary Production (GPP) Version 2.0 product. They are calculated globally and made available to the users in near-real time every 10 days. They are delivered with a maximum of 3 days lag in Near Real Time (NRT), followed by consolidations in the course of the next 6 dekads.

1 BACKGROUND OF THE DOCUMENT

1.1 SCOPE AND OBJECTIVES

The Product User Manual (PUM) is the primary document that gives an overview of the product characteristics, in terms of algorithm, technical characteristics, and main validation results. The key information is summarized in Table 1.

Table 1: Summary of key information

Input data The DMP, GDMP, NPP and GPP Version 2.0 product is derived from the CLMS Fraction of Absorbed Photosynthetically Active Radiation (fAPAR) Version 2.0, and hence follows the delivery scheme of this dataset [CGLOPS1_PUM_FAPAR300m-V2.0].	See §2.2.2 p.16
Retrieval algorithm The DMP, GDMP, NPP and GPP product is computed with a Monteith variant.	See §2.2 p.16
File naming c_gls_<PROD>[-<RTx>]_<YYYYMMDDHHMM>_<AREA>_<SENSOR>_V<VERSION>	See §3.1 p.25
File format and product content Separate multi-layer netCDF files containing DMP, GDMP, NPP or GPP variable and a QFLAG	See §3.2 and §3.3 pp. 26-26
Projection and grid information Regular lat/lon grid (plate carrée) with resolution 1/336°	See §3.4.2 p.35
Spatial information Longitude [180°W – 180°E], Latitude [85°N – 65°S]	See §3.4.2 p.35
Temporal information Three products per month, dated day 10, day 20 and last day of each month, corresponding to the end date of the dekad. The product is provided in the following consolidations: RT0 (at dekad D), RT1 (at dekad D+1), RT2 (at dekad D+2) and RT6 (at dekad D+6).	See §3.4.3 p.35

Data policies	See §3.5
Data access is free and open	p.36
Data access	See §3.6
More details on data access are available on https://land.copernicus.eu/en/getting-oriented-with-the-clms-website.	p.37
Contact	See §3.6
CLMS helpdesk: https://land.copernicus.eu/en/contact-service-helpdesk	p.37

1.2 CONTENT OF THE DOCUMENT

This document is structured as follows:

- Chapter 2 summarizes the retrieval methodology
- Chapter 3 describes the technical characteristics of the product
- Chapter 4 reports the main results of the quality assessment and validation exercise
- Chapter 5 gives insights of the product usage

The requirements are recalled in Annex.

1.3 RELATED DOCUMENTS

1.3.1 Applicable documents

AD1: Part 2: Technical specifications of Framework Service Contract – Operation of the bio-geophysical variables systematic monitoring of the Global Land Component of the Copernicus Land Service 'CGLOPS' JRC/2023/OP/0273, 19th April 2023.

Available at https://etendering.ted.europa.eu/cft/cft-display.html?cftId=13795

1.3.2 Input

Document ID	Descriptor
CGLOPS1_ATBD_DMP-NPP300m-V2.0	Algorithm Theoretical Basis Document of the Dry Matter Productivity and Net Primary Production Version 2.0

CGLOPS1_ATBD_FAPAR300m-V2.0	Algorithm Theoretical Basis Document of the fAPAR Version 2.0
CGLOPS1_PUM_FAPAR300m-V2.0	Product User Manual of the fAPAR Version 2.0
CGLOPS1_QAR_DMP-NPP300m-V2.0	Quality Assessment Report of the Dry Matter Productivity and Net Primary Production Version 2.0

These documents are available on the CLMS website: https://land.copernicus.eu/en/technical-library

1.3.3 External reference

GCOS#245 The 2022 GCOS ECVs Requirements

Available at https://library.wmo.int/records/item/58111-the-2022-gcos-ecvs-requirements-gcos-245?offset=1

2 ALGORITHM

2.1 INTRODUCTION

NPP, or Net Primary Production, is an indicator of the rate at which plants produce net useful chemical energy expressed in mass of carbon per unit area per day (gC/m²/day). Similarly, DMP, or Dry Matter Productivity, gives an indication of the dry matter biomass increase (growth rate) of the vegetation and is directly related to the NPP, but customized for agro-statistics and expressed in kilograms of dry matter (kg DM) per hectare per day.

DMP and NPP images, in this case derived from PROBA-V (from January 2014 to December 2018) and from Sentinel-3/OLCI (from January 2019 onwards) imagery combined with (modelled) meteorological data from the European Centre for Medium-Range Weather Forecasts (ECMWF), are made available at 300m resolution and are updated every 10 days.

Dry Matter Productivity (DMP) and Gross Dry Matter Productivity (GDMP), or similarly Net Primary Production (NPP) and Gross Primary Production (GPP), are two components expressing components of Primary Production of ecosystems related to the creation of new organic matter by vegetation. GDMP (GPP) is the total amount of dry matter (carbon) "fixed" by land plants per unit time through photosynthesis (atmospheric CO2 into organic compounds). A substantial fraction of GDMP (GPP) supports plant autotrophic respiration (Ra), with the remainder allocated to the Dry Matter Productivity (DMP) (or Net Primary Production, NPP) of plant structural biomass in stems, leaves, and fruit, and, to a much lesser extent, volatile organic compounds used in plant defence and signalling (Gough, et al., 2011). GDMP (GPP) and DMP (NPP) both represent the above-and belowground part of plants. To convert DMP (NPP) into the aboveground part a root to shoot ratio should be applied on the DMP (NPP) values.

The DMP and NPP products are distributed in 4 separate files, named DMP (Dry Matter Productivity), Gross Dry Matter Productivity (GDMP), Net Primary Production (NPP) and Gross Primary Production (GPP). These DMP and GDMP are in fact equivalent of, respectively, NPP and GPP. The only difference (expect for the naming) is that DMP/GDMP are expressed in kg Dry Matter per hectare per day (kg DM/ha/day) while GPP/NPP are typically expressed as gram of carbon per square meter per day (gC/m^2/day). The name and unit choice of DMP was originally designed to align with the terminology used in agricultural monitoring applications. When the DMP algorithm and product were adopted by the Copernicus Land Monitoring Service, the name and unit where adopted. But, in fact, DMP could be better called Dry Matter Production (instead of Productivity). To convert the values and units of DMP to NPP or GDMP to GPP a simple conversion factor was applied:

First, kg DM/ha/day can be converted into g DM/m^2/day since:

1 kgDM/ha/day = 1000 gDM/ha/day = 0.1 gDM/m^2/day

For the conversion of Dry Matter (DM) into Carbon (C) the conversion factor defined by IPCC (2003) can be used which states that vegetation dry matter generally consists of 50% of carbon. So gC/gDM = 0.5. To summarize, NPP and GPP are calculated from DMP and GDMP as follows:

$$NPP\ [gC/m^2/day] = DMP\ [kgDM/ha/day] \times 0.5 \times 0.1 \quad \text{(Eq. 1)}$$

$$GPP\ [gC/m^2/day] = GDMP\ [kgDM/ha/day] \times 0.5 \times 0.1 \quad \text{(Eq. 2)}$$

Similar to the relation between NPP and DMP also the agronomic equivalent of the Gross Primary Productivity (GPP) can be defined as Gross Dry Matter Productivity (GDMP). Scaling from NPP to GDMP or oppositely is similar to the scaling described above. The main difference between the NPP/DMP and GPP/GDMP is conversion with the Carbon Use Efficiency (CUE).

$$NPP = GPP \times CUE \text{ or } DMP = GDMP \times CUE$$

2.2 Retrieval Methodology

2.2.1 Background

A simplified Monteith (Monteith, 1972) approach, using remote sensing data to provide the fraction of absorbed PAR-radiation (fAPAR), was first implemented by Veroustraete (1994) and provided NPP estimates for European forests. The current version of the DMP (and associated products) is a modification of this original model, operating on meteorological data from ECMWF and the CLMS fAPAR.

2.2.2 Input data

Meteorological data: Global meteorological data are retrieved from DTN, providing for each "grid-cell" (at 0.25° resolution) the values of all standard meteorological variables. Basically, all daily data are "operational forecasts for the next 24 hours" derived from ECMWF HRES model. Radiation, T_{min} and T_{max} are derived from this daily dataset for the DMP calculations.

fAPAR corresponds to the fraction of photosynthetically active radiation absorbed by the green elements of the canopy. It depends on canopy structure, vegetation element optical properties and illumination conditions. The 300 m DMP uses a 10-daily fAPAR based on PROBA-V data (from January 2014 to December 2018) and on Sentinel-3 data (from January 2019 onwards) [CGLOPS1_ATBD_FAPAR300m-V2.0]).

Land Cover information: The DMP 300m V2.0 uses biome specific Light Use Efficiency (LUE) values. The information on the global distribution of land cover types comes from the ESA CCI Land

Cover Map (epoch 2010), which was derived from ENVISAT-MERIS and SPOT-VGT imagery of the period 2008-2010. The land cover classes are based on the UN Land Cover Classification System (LCCS). The global map is shown in Figure 1.

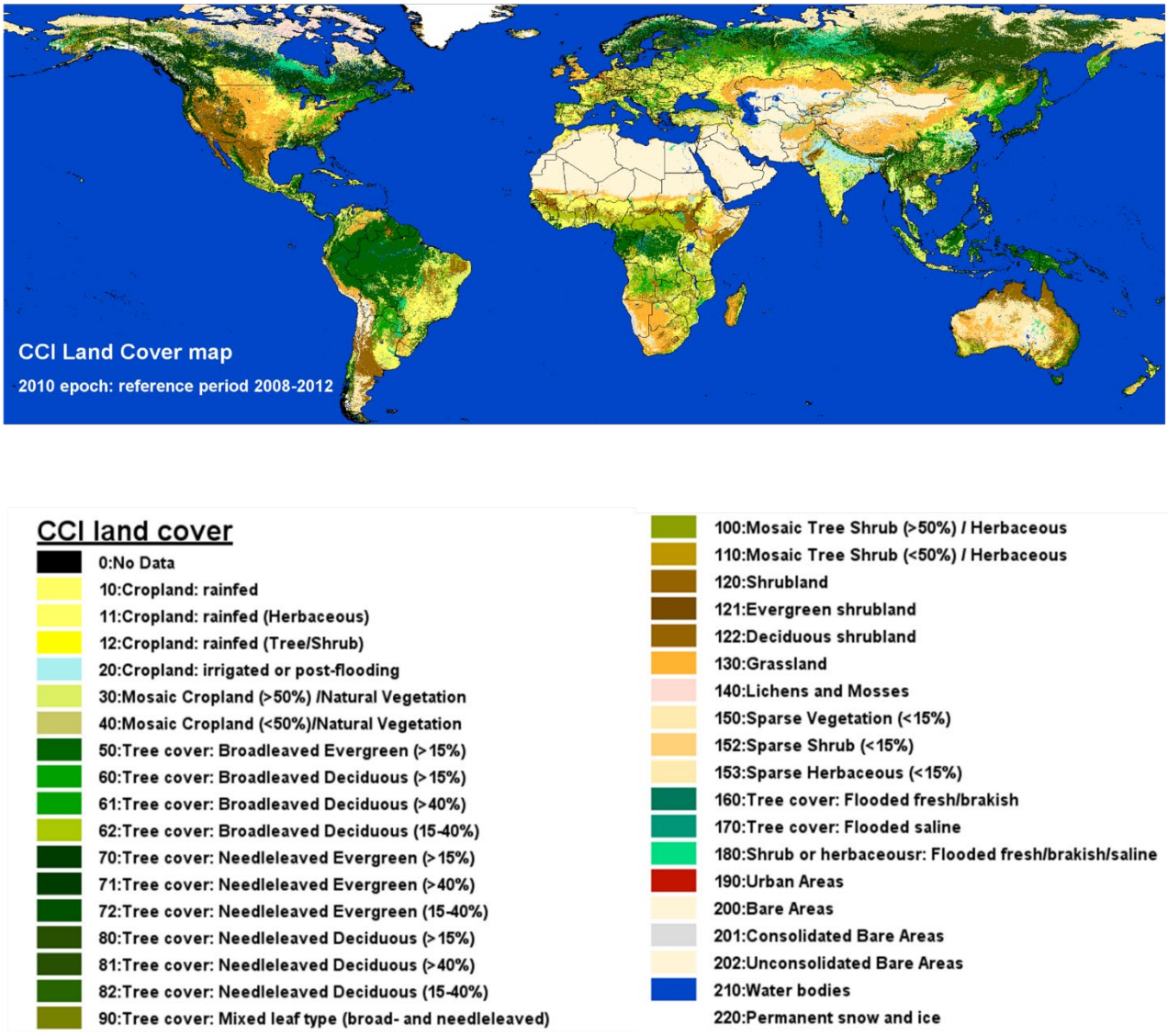


Figure 1: Global land cover map (ESA CCI epoch 2010) (top) and legend (bottom) derived from ENVISAT MERIS and SPOT-VGT data.

2.2.3 Processing steps

The input fAPAR product has small values over non vegetated areas, like deserts. This results in a positive DMP/GDMP/NPP/GPP value that are not realistic. Therefore, the low values of fAPAR are removed in the following way (see also Figure 2):

(1) All fAPAR values ≤0.05 are set to 0

(2) All fAPAR values >0.05 and <0.15 are rescaled linearly between 0 and 0.15

(3) All fAPAR values ≥0.15 remain unchanged.

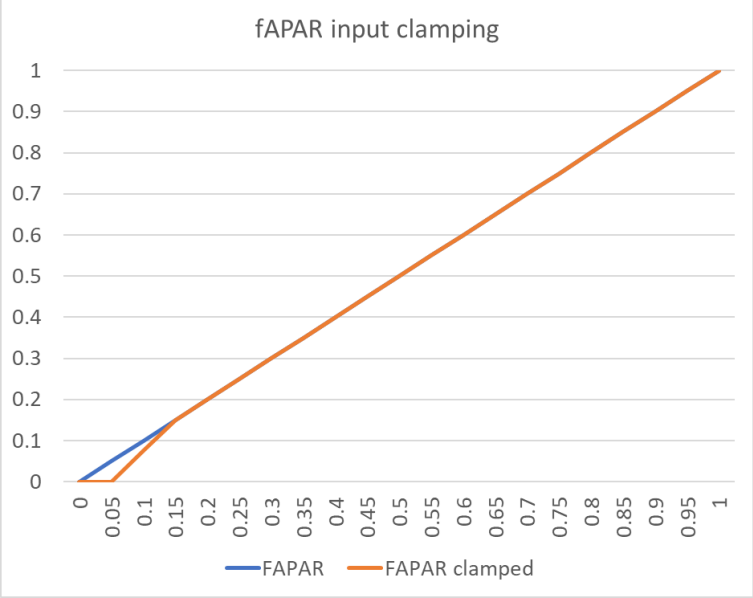


Figure 2: fAPAR clamping method.

The CLMS GDMP and DMP product is currently computed with the following Monteith variant (see Table 2):

$$GDMP = R \times \varepsilon c \times fAPAR \times \varepsilon_{LUEc} \times \varepsilon_T \times \varepsilon_{CO2}[\times\varepsilon_{RES}] \quad \text{(Eq. 3)}$$

$$DMP = R \times \varepsilon c \times fAPAR \times \varepsilon_{LUEc} \times \varepsilon_T \times \varepsilon_{CO2} \times CUE[\times\varepsilon_{RES}] \quad \text{(Eq. 4)}$$

Table 2: Individual terms in the Monteith variant used for CLMS GDMP/DMP. All terms are expressed on a daily basis.

TERM	*MEANING*	*VALUE*	*UNIT*
GDMP	Gross Dry Matter Productivity	0 - 640	kgDM/ha/day
DMP	Dry Matter Productivity	0 – 320	kgDM/ha/day
R	Total shortwave incoming radiation (0.2 – 3.0µm)	0 – 320	GJ_T/ha/day
ε_c	Fraction of PAR (0.4 – 0.7µm) in total shortwave	0.48	J_P/J_T
*f*APAR	PAR-fraction absorbed (AP) by green vegetation	0.0 ... 1.0	J_{AP}/J_P
ε_{LUEc}	Light use efficiency (DM=Dry Matter) at optimum	Biome-specific	$kgDM/GJ_{AP}$
ε_T	Normalized temperature effect	0.0 ... 1.0	-
ε_{CO2}	Normalized CO_2 fertilization effect	0.0 ... 1.0	-
CUE	Carbon Use Efficiency (GPP/NPP or GDMP/DMP ratio)	0.5	-
ε_{RES}	Fraction kept after omitted effects (drought, pests...)	1.0	-

A detailed description of the terms is given in the ATBD [CGLOPS1_ATBD_FAPAR300m-V2.0].

Given the simple elaboration of the epsilons (and dropping the factor ε_{RES}), (Eq. 3) and (Eq. 4) can be rewritten as:

$$\begin{aligned} GDMP &= fAPAR \times \varepsilon_{LUEc} \times R \times \varepsilon c \times \varepsilon_T \times \varepsilon_{CO2} \\ &= fAPAR \times \varepsilon_{LUEc} \times R \times \varepsilon(T, CO_2) \\ &= fAPAR \times \varepsilon_{LUEc} \times GDMP_{max} \end{aligned} \quad \text{(Eq. 5)}$$

$$DMP = GDMP \times CUE \quad = GDMP \times 0.5 \quad \text{(Eq. 6)}$$

with $\varepsilon(T, CO_2) = \varepsilon_c \times \varepsilon_T \times \varepsilon_{CO2}$ and ε_{LUEc} is the land cover specific LUE. This formulation better highlights the fact that (within the limits of the described model) GDMP is only determined by four basic factors: *f*APAR, radiation (R), temperature (T) and CO_2. At the same time, (Eq. 7) provides a practical method which allows bypassing the differences in temporal and spatial resolution between

the inputs. In practice, the meteorological inputs (R, T) are provided on a daily basis and at a very low resolution, here 0.25° × 0.25°. On the contrary, *f*APAR is provided on a dekadal basis, having pixels with a size of around 300x300m. The ESA CCI land cover map is also applied at 300m resolution to include the Light Use Efficiency (LUE or ε_{LUEc}) factor. Then, the final GDMP/DMP-product has this 300m resolution and dekadal frequency.

So, in practice, we use the procedure (Eerens et al., 2004) illustrated in Figure 3 and can be formulated as follows (the subscripts 1 and 10 indicate daily and dekadal products, N_d is the number of days in each dekad)

$$GDMP10 = fAPAR10 \times \varepsilon_{LUEc} \times GDMPmax,10$$
$$\text{with } GDMPmax,10 = \{\Sigma GDMPmax,1\}/Nd \quad \text{(Eq. 7)}$$

$$DMP_{10} = GDMP_{10} \times CUE \quad \text{with} \quad CUE = 0.5 \quad \text{(Eq. 8)}$$

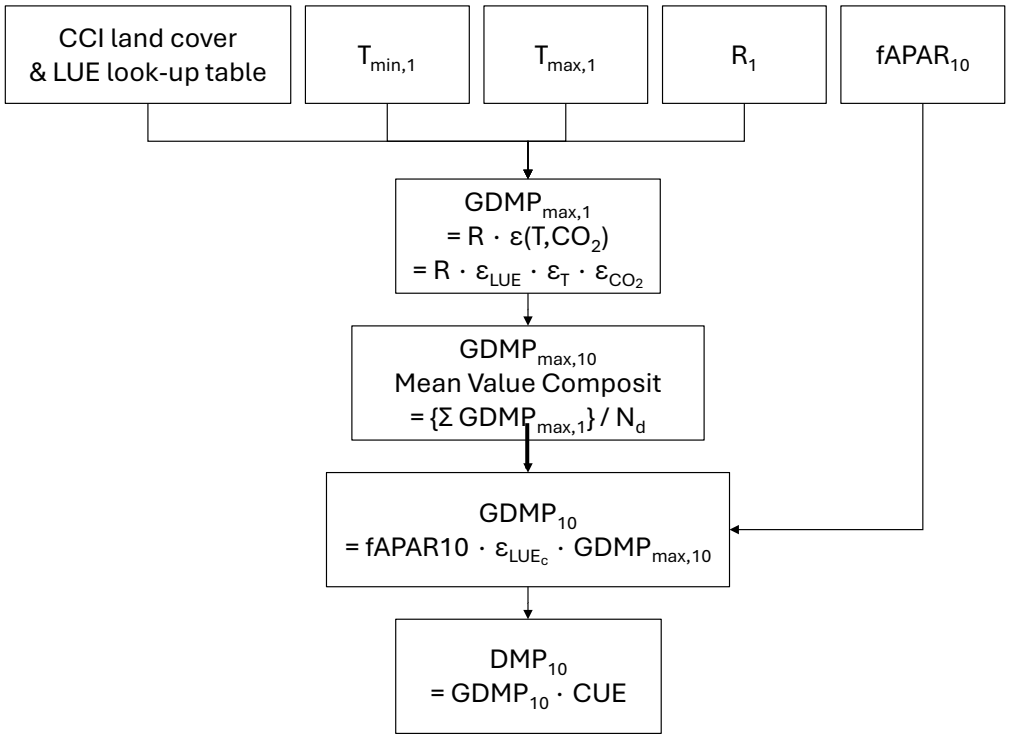


Figure 3: Process flow of GDMP/DMP. Based on meteorological data, a daily GDMPmax is estimated. At the end of each dekad, a Mean Value Composite of these GDMPmax images is calculated. At the same time, a 10-daily fAPAR product is generated. The final $GDMP_{10}$ product is retrieved by the simple multiplication of the latter two images and, then, converted to DMP_{10}.

NPP and GPP are derived from DMP and GDMP using the following formulae:

$$\text{NPP } [\text{gC/m}^2\text{/day}] = \text{DMP } [\text{kgDM/ha/day}] \times 0.5 \times 0.1 \quad \text{(Eq. 9)}$$

$$\text{GPP } [\text{gC/m}^2\text{/day}] = \text{GDMP } [\text{kgDM/ha/day}] \times 0.5 \times 0.1 \quad \text{(Eq. 10)}$$

2.2.4 Near Real Time retrieval of the DMP (NPP) 300m Version 2.0

The details of the algorithm are described in the ATBD [CGLOPS1_ATBD_FAPAR300m-V2.0].

The production of the DMP (NPP) is managed similarly to the production of fAPAR Version 2.0. A brief description is given below but the reader is also referred to the product user manual of this fAPAR [CGLOPS1_PUM_FAPAR300m-V2.0]. A distinction is made between the past-time series production and the near real time products.

The compositing for real time estimates is achieved according to the scheme described in Figure 4. We consider a particular dekadal date, *D*.

- In real time mode (RT0) (when the actual date corresponds to *D*, i.e. the first line in Figure 4), the compositing is achieved using only the past observations, with a maximum compositing window spanning c_- days in the past. The value of the product at dekadal date *D* is therefore relatively inaccurate.
- Then, the observations accumulate after dekadal date *D* when time passing. The value at dekadal date *D* is updated using the observations available after dekadal date *D* within the consolidation period. The accuracy of the product value progressively improves during this consolidation period.
- At the end of the consolidation period that lasts c_+ days, the value of the product value at dekadal date *D* have converged towards the 'historical' value, i.e. when there is at least c_- days (maximum length of the compositing window) after dekadal date *D*.

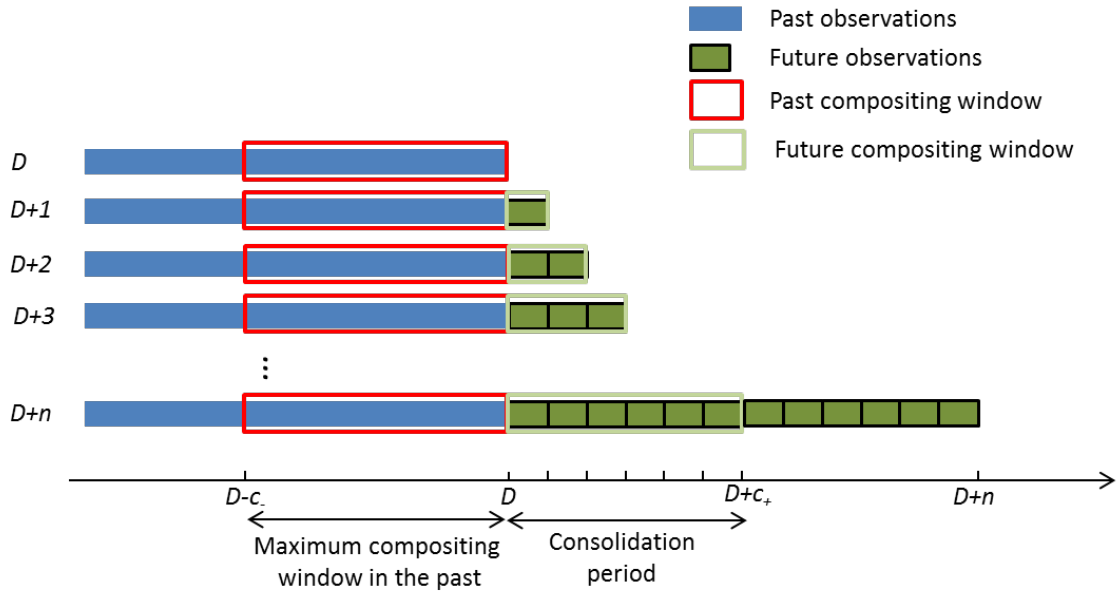


Figure 4: Scheme showing the compositing used for real time estimates

Note that the compositing period is not symmetric: the maximum compositing window in the past is longer than the maximum compositing window in the future (after date *D*) that is limited by the length of the consolidation period. The length of the maximum compositing window is set to 60 days, however more daily observations are used to classify the pixel as Evergreen Broadleaf Forest (EBF) or no-EBF. If not enough daily observations are available to estimate the dekadal value, either an interpolation with the surrounding dekads is performed or in case of EBF the value will be based on the previous dekadal value.

Once a new dekad is available after dekadal date D, the value of the product at dekadal date D is updated and replaces the previous update. This allows saving significant space. The reliability of the estimate will increase with the consolidations. The product is provided in the following consolidations: RT0 (at dekad D), RT1 (at dekad D+1), RT2 (at dekad D+2) and RT6 (at dekad D+6).

2.3 DIFFERENCES WITH THE PREVIOUS VERSION

The only difference between DMP 300m V1.0, DMP 300m V1.1, on one hand, and DMP 300m V2.0, on the other hand, is that V1.0 and V1.1 are derived from the fAPAR 300m V1.0, based upon PROBA-V data, and fAPAR 300m V1.1, based upon Sentinel-3 OLCI data, while V2.0 is derived from the fAPAR 300m V2.0 based upon PROBA-V data and the Sentinel-3 OLCI data. The differences between fAPAR 300m V1.0, fAPAR 300m V1.1 and fAPAR 300m V2.0 are described in Section 4.5 of [CGLOPS1_ATBD_FAPAR300m-V2.0].

2.4 LIMITATIONS

Following remarks should be taken into consideration when interpreting the DMP (NPP) products:

- Some potentially important factors, such as drought stress, nutrient deficiencies, pests and plant diseases, are omitted in the DMP (NPP) product. As a consequence, the product might better be called "potential" DMP or NPP. Especially for drought events, which has the strongest impact on GPP, the current DMP is expected to overestimate biomass production. Only when the adverse effects of these residual vegetation stresses (drought, diseases and shortages of nutrients) impact the RS-derived fAPAR, it will be captured by the DMP and NPP.
- The biome-specific LUE's are calibrated against flux tower Eddy Covariance data. These flux data are considered as ground truth measurements of GPP. But, in fact, this GPP is a combination of Net Ecosystem Exchange measurements and a modelled respiration factor. Additionally, there are other reasons to question the relation between flux measurements and modelled GPP estimates. The spatial representativeness of the tower, for example, is strongly dependent on local conditions like the towers footprint, wind direction, etc. Thus, caution should be made when comparing 1 km pixel values with tower observations. Each tower also has its specific environmental conditions, like soil properties, precipitation, elevation, … This makes it difficult to generalize tower specific parameterization into broad categories like land cover.
- The biome-specific parametrization requires accurate information on the land cover type. At global scale, a land cover map will always contain a fraction of misclassified pixels.
- The in-situ data availability can determine the reliability of the biome-specific LUE's. For example, for needle leaved deciduous forests, data from only one tower is available.
- The DMP algorithm does not include a water stress factor to account for short term drought stresses. In drought-sensitive vegetation types, this formulation can lead to an overestimation of the actual plant productivity. The temperature dependency factor uses generalized values derived from the parameterization on European forests (Veroustraete et al., 2002) and these parameterizations may not hold for other biome types and agro-ecological zones. Especially the difference between C3 and C4 vegetation types is pronounced. C3 plants (like the forests used in the current calibration) will have their optimal temperate around 20-25 degrees whereas C4 plants will reach their optimum around 30-35 degrees (Massad et al., 2007; Yamori et al., 2014).
- fAPAR is included in Light Use Efficiency (LUE) models such as the DMP (NPP) to represent in a relative way an observation of the actual uptake of solar radiation by plants. Combined with the modelled response of vegetation LUE to environmental conditions (represented by the meteorological data as model input) satellite based GDMP (GPP) is estimated. The assumed relation between fAPAR and GDMP (GPP) fits fairly well in general but there are a number of known cases where satellite based fAPAR is not capturing sufficiently accurate GPP or GDMP dynamics:

  - Cloud contamination can result in changes in VIs that can confound the real seasonality of the vegetation.
  - Optical sensors lack the capability to detect the full canopy activity of dense vegetation such as tropical forests.
  - Vegetation greenness as reflected by fAPAR is not able to capture seasonal photosynthetic activity of evergreen trees because of the decoupling of greenness and photosynthesis (Huang et al., 2019).
  - A similar decoupling can occur over wintergreen vegetation such as cover crops where fAPAR values are high but there is little or none photosynthetic activity and plant growth. This can result in overestimations of GDMP/DMP or GPP/NPP.
  - In general, the spectral response of vegetation canopy is insufficient to capture drought effects of GDMP or GPP. Furthermore, the spectral sensitivity of vegetation canopies differs between ecosystems. Trees which have deeper roots are most resistant to decreased soil water and will have a lower sensitivity, while the sensitivity is generally higher for cropland, grassland and shrubland (Zhang et al., 2016).
- The autotrophic respiration is calculated as a simple fraction of GDMP or GPP and is therefore assumed to have the same ecophysiological behaviour. In other models (e.g. used for MODIS product retrieval), the autotrophic respiration is considered as an independent component.
- The CO_2 fertilization factor relies on the original definition by Veroustraete (Veroustraete, 1994). We acknowledge that more sophisticated and state-of-the-art CO_2 fertilization submodels should be developed. But in the Copernicus Land Monitoring Service, the focus is on algorithm implementation rather than on fundamental research on algorithm components. Hence, we have to rely on existing described methods. If state-of-the-art modules are being developed that perform reliably better than the current approach, then it can be considered to implement them in a future version of the GDMP and GPP algorithm.

3 PRODUCT DESCRIPTION

3.1 FILE NAMING

The **DMP, GDMP, NPP and GPP** 300m products follow the naming standard:

c_gls_<Acronym>[-<RTx>]_<YYYYMMDDHHMM>_<AREA>_<SENSOR>_V<VERSION>

where

- <Acronym> is either DMP300, GDMP300, NPP300 or GPP300, the 300 references that the product is provided at 300m spatial resolution.
- <RTx> is the consolidated period, as explained in section 3.4.3:
  - RT0: Near Real Time product
  - RTn: Consolidated Real Time Product, where n equals the number of times the RT0 product was updated.
- <YYYYMMDDHHMM> gives the temporal location of the file. YYYY, MM, DD, HH, and MM denote the year, the month, the day, the hour, and the minutes, respectively.
- <AREA> gives the spatial coverage of the file. In this case, <AREA> is GLOBE, the name of the entire globe.
- <SENSOR> gives the name of the sensor used to retrieve the product, with S3 referring to Sentinel-3 and PV to PROBA-V.
- <VERSION> shows the processing line version used to generate these DMP, GDMP, NPP and GPP products. The version denoted as M.m.r (e.g. 1.1.1), with 'M' representing the major version (e.g. V1), 'm' the minor version (starting from 0) and 'r' the production run number (starting from 1) (Table 3).

Table 3: Explanation in version numbering and recommendations for using efficiently the products

Versions	Differences	Recommendations
Major	Significant change to the algorithm.	Do not mix various major versions in the same applications, unless it is otherwise stated.
Minor	Minor changes in the algorithm	Can be mixed in the same applications, but require attention or modest modifications
Run	Fixes to bugs and minor issues. Later run automatically replaces former	Consider it as a drop-in replacement

3.2 FILE FORMAT

The DMP/GDMP/NPP/GPP Version 2.0 products are distributed in compressed Network Common Data Form version 4 (netCDF4) files with attributes compliant with version 1.6 of the Climate and Forecast (CF) conventions. Separate products are delivered for all variables.

The multi-layer GDMP netCDF files contain the following layers:

- GDMP: Gross Dry Matter Productivity value
- QFLAG: Quality Flag indicator

The multi-layer DMP netCDF files contain the following layers:

- DMP: Dry Matter Productivity value
- QFLAG: Quality Flag indicator

The multi-layer GPP netCDF files contain the following layers:

- GPP: Gross Primary Production value
- QFLAG: Quality Flag indicator

The multi-layer NPP netCDF files contain the following layers:

- NPP: Net Primary Production value
- QFLAG: Quality Flag indicator

3.3 PRODUCT CONTENT

3.3.1 Data Files

Table 4 gives the digital numbers associated with the physical DMP, GDMP, NPP and GPP range and no data values. Note that no uncertainties are associated to DMP, GDMP, NPP and GDMP values. The physical values are retrieved by:

$$PhyVal = DN \times scale_factor + add_offset \quad \text{Eq. 11}$$

where DN is the digital number in the file and the scale_factor and add_offset are given in the Table 4. This rescaling formula does not apply to the binary numbers representing special flag values.

Table 4: Range of values and scaling factors of GDMP/DMP/GPP/NPP.

	GDMP (kg Dry Matter / ha / day)	**DMP** (kg Dry Matter / ha / day)	**GPP** (g C / m² / day)	**NPP** (g C / m² / day)
Minimum physical value	0	0	0	0
Maximum physical value	655.34	327.67	32.767	16.3835
Maximum DN value	32767	32767	32767	32767
Missing value	-1	-1	-1	-1
Specific value for sea	-2	-2	-2	-2
Scale_factor	0.02	0.01	0.001	0.0005
Add_offset	0	0	0	0

Once properly decoded, the physical DMP (GDMP) values range between 0 and 327.67 (655.34) kg dry matter per hectare per day, and the physical NPP (GPP) values range between 0 and 16.3835 (32.767) kg carbon per square meter per day. The lowest values are found over non-vegetated areas like deserts, urban areas or outside the growing season in cold climate zones. The highest values indicate high productive pixels, often found over forested areas in the peak of the growing season or in tropical zones.

3.3.2 QFLAG image

In addition to the quality flags in the negative range of the data image (Table 4), the quality of the GDMP, DMP, GPP and NPP product is also directly linked to the quality of the input fAPAR.

Table 5 provides the DMP, GDMP, NPP, GPP 300m version 2.0 QFLAG layer. This layer is derived from the fAPAR 300m V2.0 quality indicator [CGLOPS1_PUM_FAPAR300m-V2.0] as referenced in the last column of Table 5.

Table 5: Range of values of QFLAG.

Bit	Qualitative indicator	Meaning	Mapping on fAPAR300
1	fAPAR missing or invalid	0 = Available fAPAR300 value 1 = Missing fAPAR300 value	fAPAR = 255
2	Meteorological data missing or invalid	0 = Available Meteorological values 1 = Missing Meteorological values	-
3	Not used		
4	Evergreen Broadleaf Forest (EBF)	1 = EBF 0 = no EBF	QFLAG bit 2
5	Not used		
6	Low fAPAR correction	1 = fAPAR<0.15 and rescaled according to Figure 2 0 = fAPAR is unchanged	
7	fAPAR computation for EBF	0 = fAPAR from daily observations 1 = fAPAR filled from previous dekad	QFLAG bit 5
8	fAPAR gap filling by interpolation	0 = fAPAR not interpolated 1 = fAPAR interpolated	QFLAG Bit 3

3.3.3 File and Layer attributes

The netCDF file contains a number of netCDF metadata attributes

- on the file-level (Table 6)
- on layer level (Table 7 and Table 8)
- at the level of the standard dimension variables for latitude (‘lat’) and longitude (‘lon’), holding one value per row or column respectively (Table 9)
- at the level of the grid mapping (spatial reference system) variable (‘crs’) (see Table 10)

Table 6: Description of NetCDF file attributes

Attribute name	Description	Data Type	Example
Conventions	Version of the CF-Conventions used	String	CF-1.6
title	A description of the contents of the file	String	10-daily Dry Matter Productivity 333M: GLOBE 2023-09-10T00:00:00Z
institution	The name of the institution that produced the product	String	VITO NV
source	The method of production of the original data	String	Derived from EO satellite imagery
history	A global attribute for an audit trail. One line, including date in ISO-8601 format, for each invocation of a program that has modified the dataset.	String	2025-09-26: Processing line DMP
references	Published or web-based references that describe the data or methods used to produce it.	String	https://land.copernicus.eu/global/products/dmp
archive_facility	Specifies the name of the institution that archives the product	String	VITO
product_version	Version of the product (VM.m.r)	String	V2.0.1
time_coverage_start	Start date and time of the total coverage of the data for the product.	String	2023-09-01T00:00:00Z
time_coverage_end	End date and time of the total coverage of the data for the product.	String	2023-09-10T23:59:59Z
platform	Name(s) of the orbiting platform(s)	String	SENTINEL3A,SENTINEL3B
sensor	Name(s) of the sensor(s) used	String	OLCI
identifier	Unique identifier for the product	String	urn:cgls:global:dmp300_v2_333m:DMP300_202309100000_GLOBE_S3_V2.0.1
parent_identifier	Identifier of the product collection (time series) for the product in CLMS metadata catalogue.	String	urn:cgls:global:dmp300_v2_333m

Attribute name	Description	Data Type	Example
long_name	Extended product name	String	Dry matter productivity Net Primary Production
orbit_type	Orbit type of the orbiting platform(s)	String	LEO
processing_level	Product processing level	String	L3
processing_mode	Processing mode used when generating the product (Near-Real Time, Consolidated or Reprocessing)	String	Near Real Time
copyright	Text to be used by users when referring to the data source of this product in publications (copyright notice)	String	Copernicus Service information 2025

Table 7: Description of NetCDF layers attributes for DMP/GDMP

Attribute name	Description	Data Type	Example
CLASS	Dataset type		
standard_name	A standardized name that references a description of a variable's content in CF-Convention's standard names table. Note that each standard_name has corresponding unit (from Unidata's udunits).	String	Dry_matter_ productivity
long_name	A descriptive name that indicates a variable's content. This name is not standardized. Required when a standard name is not available.	String	Dry matter productivity 333M
scale_factor	Multiplication factor for the variable's contents that must be applied in order to obtain the real values. Omit in case the scale is 1.	Float	0.01
add_offset	Number to be added to the variable's contents (after applying scale_factor) that must be applied in order to obtain the real values. Omit for offset 0.	Float	0.0
units	Units of a the variable's content, taken from Unidata's udunits library. Empty or omitted for dimensionless variables. Fractions should be indicated by scale_factor.	String	kg / hectare / day
valid_range	Smallest and largest values for the variable.	Integer	0, 32767

	Missing data is to be represented by one or several values outside of this range.		
_FillValue	Single value used to represent missing or undefined data and to pre-fill memory space in case a non-written part of data is read back. Value must be outside of valid_range.	Integer	-1
missing_value	Single value used to represent missing or undefined data, for applications following older versions of the standards. Value must be outside of valid_range.	Integer	-1
grid_mapping	Reference to the grid mapping variable	String	Crs
flag_values	Provides a list of the flag values. Used in conjunction with flag_meanings.	Integer	-2
flag_meanings	Descriptive phrases for each flag value, separated by blanks	String (blank separated list)	Sea

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Table 8: Description of netCDF layer attributes for NPP/GPP

Attribute	Description	Data Type	Example for NPP layer
CLASS	Dataset type		
standard_name	A standardized name that references a description of a variable's content in CF-Convention's standard names table. Note that each standard_name has corresponding unit (from Unidata's udunits).	String	Net_Primary_Production
long_name	A descriptive name that indicates a variable's content. This name is not standardized. Required when a standard name is not available.	String	Net Primary Production 333M
scale_factor	Multiplication factor for the variable's contents that must be applied in order to obtain the real values. Omit in case the scale is 1.	Float	0.0005
add_offset	Number to be added to the variable's contents (after applying scale_factor) that must be applied in order to obtain the real values. Omit for offset 0.	Float	0.0
units	Units of the variable's content, taken from Unidata's udunits library. Empty or omitted for dimensionless variables. Fractions should be indicated by scale_factor.	String	g / m² / day
valid_range	Smallest and largest values for the variable. Missing data is to be represented by one or several values outside of this range.	Integer	0, 32767
_FillValue	Single value used to represent missing or undefined data and to pre-fill memory space in case a non-written part of data is read back. Value must be outside of valid_range.	Integer	-1
missing_value	Single value used to represent missing or undefined data, for applications following older versions of the standards. Value must be outside of valid_range.	Integer	-1
grid_mapping	Reference to the grid mapping variable	String	Crs
flag_values	Provides a list of the flag values. Used in conjunction with flag_meanings.	Integer	-2
flag_meanings	Descriptive phrases for each flag value, separated by blanks	String (blank separated list)	Sea

Table 9: Description of netCDF attributes for coordinate dimensions (latitudes and longitudes)

Attribute	Description	Data Type	Example
CLASS	Dataset type	String	DIMENSION_SCALE
DIMENSION_LABELS	Label used in netCDF4 library	String	lon
NAME	Short name	String	lon
standard_name	A standardized name that references a description of a variable's content in CF-Convention's standard names table. Note that each standard_name has corresponding unit (from Unidata's udunits).	String	longitude
long_name	A descriptive name that indicates a variable's content. This name is not standardized. Required when a standard name is not available.	String	longitude
units	Units of the variable's content, taken from Unidata's udunits library.	String	degrees_east
axis	Identifies latitude, longitude, vertical, or time axes.	String	X
_CoordinateAxisType	Label used in GDAL library	String	Lon

Table 10: Description of netCDF attributes for the grid mapping variable

Attribute	Description	Data Type	Example
GeoTransform	Six coefficients for the affine transformation from pixel/line space to coordinate space, as defined in GDAL's GeoTransform	String	-180.0000000000 0.0029761905 0.0 80.0000000000 0.0 -0.0029761905
_CoordinateAxisTypes	Label used in GDAL library	String, blank separated	GeoX GeoY
_CoordinateTransformType	Type of transformation	String	Projection
grid_mapping_name	Name used to identify the grid mapping	String	latitude_longitude
inverse_flattening	Used to specify the inverse flattening (1/f) of the ellipsoidal figure associated with the geodetic datum and used to approximate the shape of the Earth	Float	298.257223563
long_name	A descriptive name that indicates a variable's content.	String	coordinate reference system
longitude_of_prime_meridian	Specifies the longitude, with respect to Greenwich, of the prime meridian associated with the geodetic datum	Float	0.0
semi_major_axis	Specifies the length, in metres, of the semi-major axis of the ellipsoidal figure associated with the geodetic datum and used to approximate the shape of the Earth	Float	6378137.0
spatial_ref	Spatial reference system in OGC's Well-Known Text (WKT) format	String	GEOGCS["WGS 84",DATUM["WGS_1984",… AUTHORITY["EPSG","4326"]]

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3.4 PRODUCT CHARACTERISTICS

3.4.1 Projection and grid information

The product is displayed in a regular latitude/longitude grid (plate carrée) with the ellipsoïd WGS 1984 (Terrestrial radius=6378km). The resolution of the grid is 1/360°. The reference is the centre of the pixel (integer degrees are in the centre of a pixel).

3.4.2 Spatial information

The Sentinel 3 derived products are provided from longitude 180°W (at centre of pixel) to 180°E and latitude 80°N to 65°S. The Proba-V derived products have only valid data between latitude 75°N and 65°S because of the sensor range.

The meteorological input (section 2.2.2) is bound from 75°N to 50°S. Hence, the land areas outside this window are filled with Missing values (Table 4).

3.4.3 Temporal information

The DMP, GDMP, NPP, GPP 300m are near real time products, updated every 10 days within a consolidated period. The temporal information "YYYYMMDDHHMM" in the filename is the "**nominal**" or reference date corresponding to the product value, (D), the last day of the dekad (10-day period), similar to the corresponding FAPAR300 input product [CGLOPS1_PUM_FAPAR300m-V2.0]. The RT0 product is updated with two consolidation periods (RT1 and RT2) and a final consolidation (RT6), as shown in Figure 5.

For example, the file c_gls_DMP300-RT0_202307310000_GLOBE_S3_V2.0.1.nc corresponds to the near real time DMP product of the third dekad (D) of July 2023, from 21/07/2023 to 31/07/2023. The input fAPAR product is generated using the past observations till D within the compositing window.

For example, the file c_gls_DMP300-RT2_202307310000_GLOBE_S3_V2.0.1.nc corresponds to the consolidated DMP product two dekads (D+2) after the third dekad of July 2023. The actual product date still corresponds to the third dekad of July 2023. However, the input fAPAR product is generated using the past observations till D within the compositing window plus the observations of the two dekads in the consolidation period (between D and D+2).

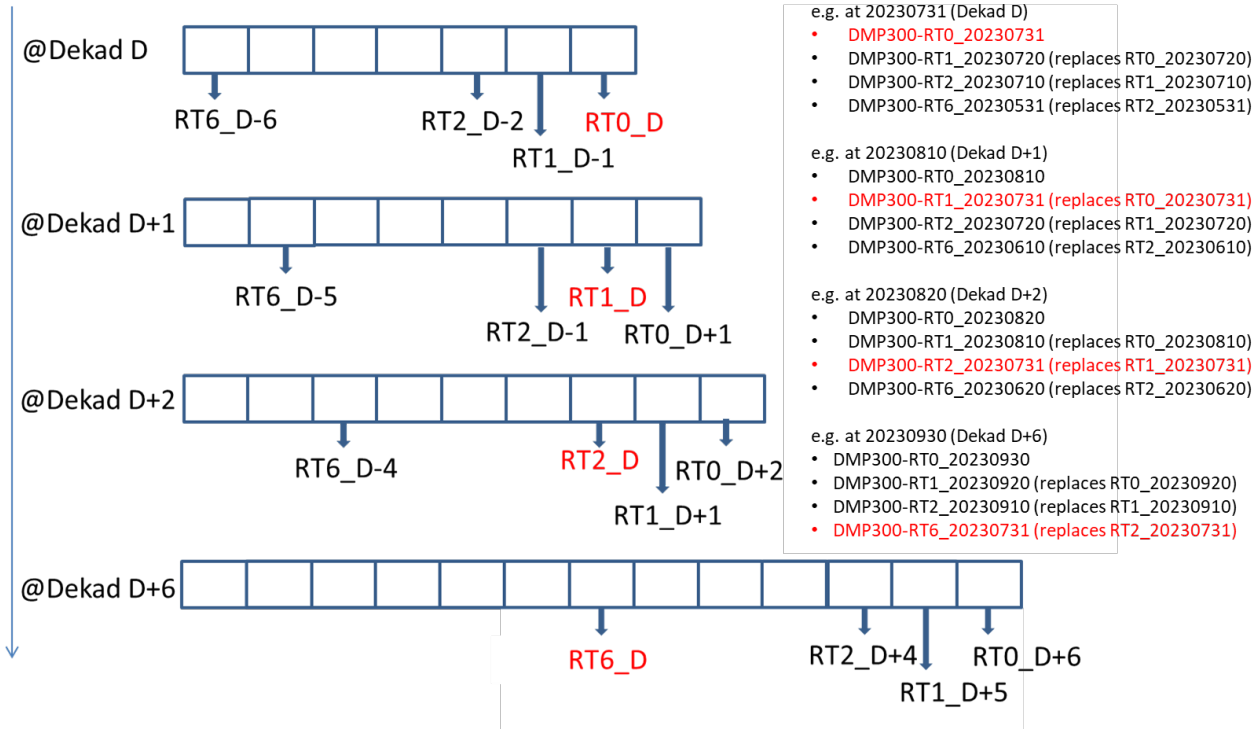


Figure 5: Consolidation period example

As the product value after the second consolidation remains stable, it has been decided not to distribute the 3rd, 4th and 5th consolidation (RT3, RT4 and RT5), but only a final (RT6) consolidation.

For reprocessed historical data, only the most accurate consolidation RT6 is produced.

3.5 DATA POLICIES

The Copernicus component is governed by Regulation (EU) No 2021/696 of the European Parliament and of the Council of 28 April 2021 establishing the Union Space Programme and the European Union Agency for the Space Programme.

The Copernicus land monitoring products and services are made available on a principle of **full, open and free access**, as established by the Commission Delegated Regulation (EU) No 1159/2013 of 12 July 2013.

How to cite and reference the products and services of the Copernicus Land Monitoring Service is explained here: https://land.copernicus.eu/en/data-policy.

3.6 ACCESS AND CONTACTS

The DMP, GDMP, NPP and GPP 300m products are available through the Copernicus Land Monitoring Service website at the address: https://land.copernicus.eu/ with contact information (helpdesk) on https://land.copernicus.eu/en/contact-service-helpdesk.

4 SUMMARY OF VALIDATION EXERCISE

The validation is ongoing.

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5 PRODUCT USAGE

5.1 TYPICAL USE

The DMP/GDMP/NPP/GPP products are provided in a single image and are commonly visualized through specialized software. Note that, in order to retrieve actual DMP, GDMP, GPP and/or NPP values from the products, the digital numbers have to be scaled using scale factors as listed in Table 4. Missing values due to clouds or snow/ice and water are indicated with special flags (see also Table 4).

These visual representations of the Dry Matter Productivity or Net Primary Production product can then be inspected to identify zones of high or low productivity and as such they are useful for general evaluations of the ecosystem productivity of an area.

5.2 ADVANCED USAGE: HOW TO ANALYZE THE DATA

5.2.1 Carbon fluxes

Quantifying carbon fluxes and stocks is essential for better understanding the global carbon cycle and improving projections of the carbon-climate feedback. Gross Primary Production (GPP) is the total carbon fixed by plants via photosynthesis and provides a key metric for understanding ecosystem productivity (Figure 6). Long-term time-series datasets (multi-decadal, global, coarse resolution) are central to climate-trend analysis, carbon cycle modelling, and ecosystem monitoring.

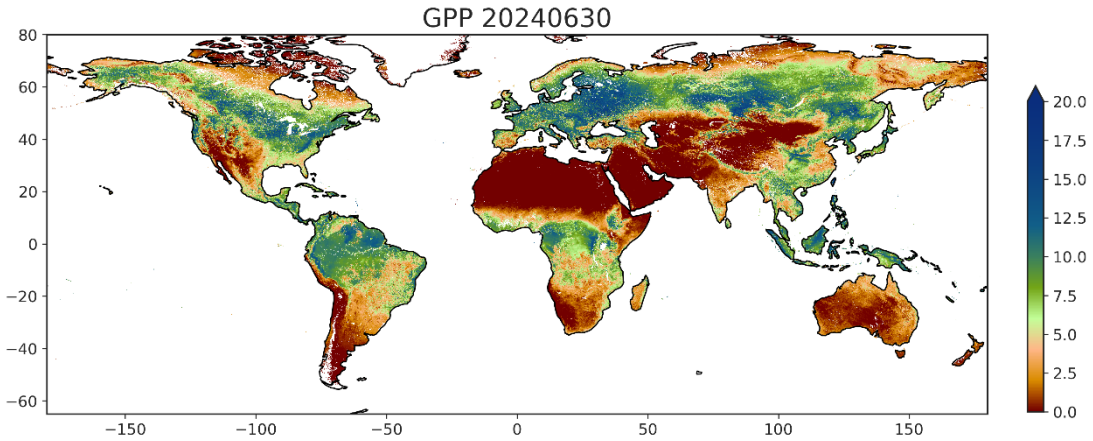


Figure 6 : Global GPP map of the last dekad of June 2024. GPP is expressed in $gC/m^2/day$.

Monitoring carbon stock changes over time using trend analysis can be used to determine which ecosystems and processes drive changes in carbon fluxes (Friedlingstein et al., 2020; Liu et al., 2015; Xiao et al., 2019). About one third of total GPP by terrestrial plants occurs in tropical forests, and

precipitation controls carbon uptake in more than 40% of vegetated land (Beer et al., 2010). However, for example El Niño events lead to reduced GPP in tropical forests of eastern Amazonia and central Africa (Malhi et al., 2018).

5.2.2 Ecosystem and vegetation dynamics

A significant portion of GPP (up to half) is lost almost immediately through autotrophic respiration, which provides energy for plant maintenance and growth. The remaining carbon after respiration is called Net Primary Production (NPP), which reflects short-term carbon storage (Valentini, 2013).

Both GPP and NPP are essential metrics for understanding terrestrial ecosystem functioning. GPP represents the total carbon fixed via photosynthesis and responds rapidly to short-term environmental drivers such as radiation, temperature, and moisture. As such, it can be analysed to capture fast ecosystem processes, such as drought stress or phenological shifts. GPP aligns strongly with direct photosynthetic indicators like solar-induced fluorescence (SIF) (Damm et al., 2015; Liu et al., 2025), making it suitable for tracking dynamic vegetation responses across ecosystems.

NPP represents carbon available for growth and long-term biomass accumulation and is therefore of interest for analysis of long-term ecosystem development and productivity trends. By combining NPP with data on heterotrophic respiration, land use change, and disturbances (e.g. wildfire), it is possible to identify net carbon sinks (with positive net ecosystem productivity, NEP) or net carbon sources (negative NEP) at regional and global scale (Wang et al., 2025).

5.2.3 Agricultural monitoring

For agronomic purposes, the Gross Dry Matter Production (GDMP) [kgDM/ha/day] is used, which is a rescaling of GPP [gC/m²/day] by taking into account the conversion between carbon and dry matter from (Penman, 2003). DMP is derived from GDMP as the net biomass increment resulting from photosynthesis after subtracting autotrophic respiration and losses. As such, DMP provides the overall growth rate of dry biomass increase of vegetation and is widely used in agricultural monitoring because it correlates with biomass accumulation, crop performance, and yield.

Global cropland use intensity can be analysed based on DMP (or NPP), inputs of fertilizer and irrigation, and biomass outputs (Niedertscheider et al., 2016). Multi-temporal analysis of DMP (or NPP) products reveals changes in crop growth during the growing season by comparing consecutive images and highlights multi-year differences when compared to past seasons or long-term averages.

For crop yield assessment, the amount of net biomass increment from the early growing stage until harvest time is accumulated. When properly calibrated against official yield statistics DMP products may be analyzed for crop yield estimation and forecasting. Multiple regression (e.g. Baruth et al., 2008; Blatchford et al., 2019) and neural networks (e.g. Zhu et al., 2021) are commonly used techniques to relate DMP and official yields. For example, NPP or DMP was used to estimate paddy

rice yield (Peng et al., 2014) or to assess grassland biomass production or carrying capacity (de Leeuw et al., 2019; Frank et al., 2025).

For example, Action Against Hunger (ACF) integrates the CLMS DMP product in their BioGenerator tool to produce annual biomass production maps and derived anomaly indicators over the rainy season in the Sahel (see example for the 2025 season in Figure 7). These derived products feed into regular biomass production bulletins[1] and can be consulted in an online geo-information viewer[2].

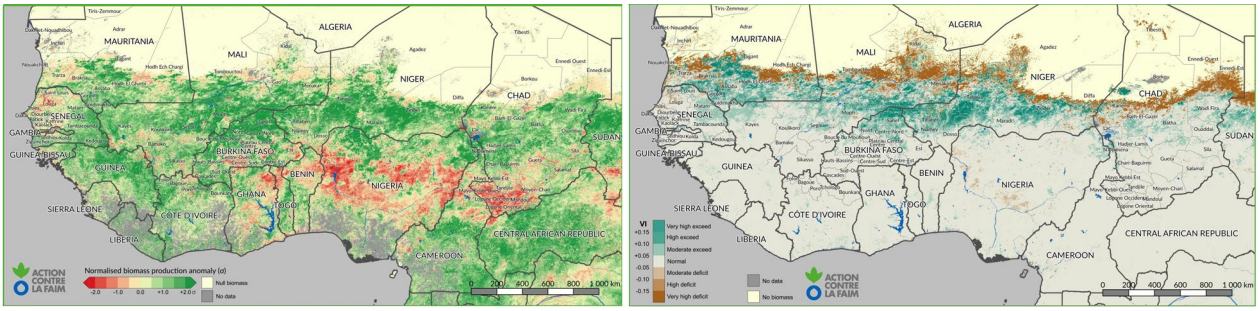


Figure 7: Normalized biomass production anomaly (left) and biomass vulnerability index (right) for the year 2025 (ACF, 2025)

[1] https://sigsahel.info/
[2] https://geosahel.info/Viewer.aspx?map=West_Africa

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7 ANNEX: USER REQUIREMENTS

The user requirements relevant for DMP and NPP are according to the applicable document [AD1].

7.1 SPECIFIC TECHNICAL DETAILS AND REQUIREMENTS

PRODUCT SPECIFICATION	
Geometric Properties:	
Baseline dataset pixel resolutions	300m
Target baseline location accuracy	Better than 0.5 pixels
Coordinate position	Centre of the pixel
Geodetical datum	WGS84
Geographic projection	Regular latitude/longitude grid
Geographic Coverage:	Global
Temporal resolution	10-day period (dekad: days 1-10, 11-20, 21 end of month)
Timeliness	Within 2 days (optimally 1 day) after the end of each dekad

DMP – DRY MATTER PRODUCTIVITY	
Amount (weight) of dry matter (DM) produced per surface unit and per time unit expressed in kilograms of dry matter per hectare per day (kgDM/ha/day). The accumulation over time leads to the assessment of the biomass (in this context, phytomass), and can be defined as above-ground mass per unit area of living plant material (GCOS#245).	
Uncertainty (2-sigma)	Threshold: 20% (can be biome specific: <10% in crop areas; 20% for grassland); Goal: 10% (can be biome specific: <10% in crop areas; 20% for grassland) [kg DM/ha/day]
Stability (per decade, i.e. 10 years)	Threshold: 10% Goal: 5%

NPP – ABOVE GROUND NET PRIMARY PRODUCTION	
Net primary production is the rate at which plants produce net useful chemical energy expressed in mass of carbon per unit area per day (gC/m²/day)	
Uncertainty (2-sigma)	Threshold: 20% (can be biome specific: <10% in crop areas; 20% for grassland); Goal: 10% (can be biome specific: <10% in crop areas; 20% for grassland);
Stability (per decade, i.e. 10 years)	Threshold: 10% Goal: 5%

7.2 FURTHER REQUIREMENTS

7.2.1 Output product composition

Products may contain various information layers and ancillary information – the base reference for product packages are the operational products as on 01.03.2023.

7.2.2 Data Structure

Data coding[3] shall be compatible with the Global Land products as on 01.03.2023 and/or follow the INSPIRE specifications, where applicable.
Ancillary information shall be as currently used and include at least the following:

- The number of measurements per pixel used to generate any synthesis product
- The per-pixel date of the individual measurement or the start-end dates of the period covered
- Quality indicators, with explicit per-pixel identification of the cause of anomalous parameter result.

The product naming and filename conventions that are used in the Copernicus Global Land component production as on 01.03.2023 shall be followed. This may be adapted for complete product collections upon agreement with the contracting authority during the Framework.

7.2.3 Data format

To ensure interoperability with the current Global Land component (operational product data formats and archive data formats) and other Copernicus services, all datasets shall be available in NETCDF. Additional format such as Cloud Optimized Geotiff (COG) or ZAR format can be proposed for production or could be requested by the Contracting Authority during the Framework Contract.

7.2.4 Uncertainties and Validation

Uncertainties indicated in the product specifications above follow the threshold and goal proposed by GCOS#245.

Uncertainties estimates should account for the error propagation uncertainty coming from input data though the retrieval algorithms as during the contract period, ESA plans to imbed uncertainties in the Sentinel-3 ground segment products and, for Sentinel-2, there is an offline tool to determine Evel 1 uncertainties; these can be used for propagation in the production chain.

Validation of the products shall conform to at least the CEOS LPV standards. Wherever appropriate the bio-geophysical variables shall be validated and compared to CEOS CAL/VAL data sets and/or

[3] Data coding is the provision for the number of bits, range of values, usage of reserved values, content of status map, conversion to physical values formulas.

Ground-Based Observations for Validation (GBOV) of Copernicus Global Land component when biophysical parameters are available.

7.2.5 Input data

Copernicus sentinel data are available from https://dataspace.copernicus.eu

Bio-geophysical variable products should be based on common base reflectance data:

- Sentinel 3: Derived Top-of-Canopy reflectance may be brokered or produced as it is the case on 01.03.2023 under the CGLOPS contracts;
- Sentinel 2: the Global Land Sentinel-2 Global Mosaic (S2GM)[4] component provides temporal mosaic of surface spectral bands that can be brokered and/or directly used.

Up to and including 2019, products archives of the Global Land component have been based on SPOT VGT, Proba-V, ENVISAT, MODIS, TOPEX/Poseidon, Jason-1, Jason-2, Jason-3, datasets, which are available through the https://dataspace.copernicus.eu.

Ancillary satellite data that is purchased through Copernicus and put at disposal of the Services is available through the Data Warehouse and will become available on the Copernicus Dataspace Ecosystem.

Ancillary data sets, other than satellite imagery described above, that might be required shall be the responsibility of the contractor.

7.2.6 Product delivery

Products shall be delivered to the Copernicus Land Component dissemination.

The Copernicus Data Space Ecosystem infrastructure will be used to provide access to the final map products.

[4] https://land.copernicus.eu/en/products/global-image-mosaic